

Original and Revised Data Collection Process

Our project changed a couple of times before reaching the final dataset. We originally wanted to create a gym equipment usage tracker, but after talking through the logistics, we realized that we would have to spend a lot of time observing different machines, days, and time periods to collect enough data. We also worried that collecting only a small number of observations would make the results too dependent on when we would be at the gym. So, we decided to switch to restaurant menu data, Because there we could collect more observations consistently from public restaurant information.

After choosing the restaurant idea, our first idea was still very vague and we didnt really know exactly the direction we wanted to go in.. We created a set of attributes and collected a small beginning sample of around 10 items. That allowed us to see that some fields were not as straightforward as expected. For example, food category, dietary type, and main ingredient could be interpreted differently by each person, and serving size information was not always shown in the same way. We updated the data dictionary, standardized units and labels, clarified what counted as one observation, and talked about cases that were a little unclear before continuing. We also thought about additional attributes during this stage of our project, but removed or avoided ones that could not be collected consistently across all restaurants.

Our method for sampling also changed. At first, we were just trying to get as many rows as we could but we later realized that taking all these items from one restaurant or one food category could bias the comparisons. So we tried to keep the number of items per restaurant more balanced and included different categories such as burgers, salads, breakfast items, desserts, bowls, and sandwiches. Because both group members collected data separately, we would throughout the data collection compare entries to make sure the same units and methods and style were being used.

The domain questions changed with the dataset as well. Our questions in the beginning were broad, like we were comparing nutrition across food categories or vegetarian and non vegetarian items. Once we looked at the actual data, we saw that some comparisons were affected by uneven category sizes and did not really give us much information. So we went toward questions that were better supported by the data. Like differences in nutrition across restaurant types, the relationship between serving size and calories, price versus protein, and high sugar in certain food categories. This process helped us move from a broad project idea to a more consistent dataset and a set of questions that showed the patterns and limitations we actually saw.